PAPER DRAFT

Probabilistic Forecasting with HMM: Use-Case in Household Electric Power Load Forecasting

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Hidden Markov Models (HMM) have been successfully applied in speech recognition, until it was replaced by the uprising of machine learning methods. Its stochastic design offers however a computational efficient way to obtain probabilistic forecasts, which this paper aims to investigate. Households electric power load profiles are chosen as application. The goal of this paper is to model the electric power load and its characteristics of a single household with HMM, and evaluate its resulting probability prediction.

Context: This paper draft is submitted for the PhD-Day of IEWT 2025. The research presented here is still in progress and not in a publishable state yet.

Note for Reviewer: Experiments of this paper are still ongoing, thus no conclusive results are presented here. A few chapters of the paper are written out in detail, while others are summarized in comprehensive bullet points.

I. INTRODUCTION

The Hidden Markov Model (HMM) is a well studied stochastic Process, which evolved around the 1980-ties laying the basis of early Speech Recognition, with its most popular publication in Rabiner(1989). With the rise of advanced machine learning methods it moved out of the focus of recent research. However, with its simple but effective structure driven by the Markov Property in its core, it still finds applications in decoding and recognition. In this paper, our aim is to investigate the use of HMM in forecasting, which has been rather a overseen utilization in the literature. In Hassan (2005) HMM was used to predict the stock market, whereas Ghasarian (2022) computed wind forecasts. In our surprise, both of these papers, as well as other investigated papers, restrict themselves on relative small HMM Models (number of hidden States ≤ 10).

Electric power load profiles of single households are chosen for the application to test HMM as a forecasting method. Its strong stochastic characteristic holds many repetitive patterns like households specific habits, plus the usage of HMM can be interpreted in a meaningful way: While the hidden Markov Chain can be represented by the unobservable process of people living their daily life, their behavior emit an measurable output in form of electric power demand, which serve as observations of the HMM model. Furthermore, the research topic of energy forecasting is gaining more and more importance in the perspective of energy transition. Haben und Co (2021) critize in their comprehensive analysis of low voltage energy forecast literature a wide oversight of the probabilistic behavior electric power load in many of the reviewed papers. Literature, which considers its probabilistic characteristic, widely uses prediction intervals or quantiles estimation, and only a handful of papers develop a framework to forecast complete probability distributions due to its computational complexity. HMM offers here a novel opportunity, because its stochastic design implies already a probabilistic method and can further compute distribution forecasts straight forward and very efficiently.

This paper researches the question, if HMM can be applied for probability forecasting in the area of load forecasting. It investigates the optimal setting of the hyperparameter space and feature tuning in order to model the characteristics of households electric power load, like seasonality and daytime differences.

Since this work is still in progress, no final results can be presented. Preliminary experiments address the problems of overfitting and indicate the limitations of HMM in probabilistic load forecasting.

The structured of this paper is as follows. Chapter 2 lays the theoretic foundation of Hidden Markov Models and explains the forecasting method. The application of HMM to power load data is introduced in chapter 3, while chapter 4 presents the proposed model design and applied data set. In chapter 5 the experiments and benchmark are conducted, and the results are discussed. The results and outlook are summarized in chapter 6.

II. HMM AS FORECASTING METHOD

HMM are driven by a Markov chain in its core, which is described by the Markov Property. The special characteristic of HMM is that this Markov process cannot be directly observed, thus "hidden". However, in each time instance an observation is emitted with a certain probability depended on the current state of the hidden Markov Chain. This chapter explains the theory of HMM as a forecasting method, for a detailed introduction to HMM see Rabiner(1989). In this paper we utilize a discrete HMM, meaning state and observation spaces are discrete and finite.

A. HMM Foundations

1. Notation

- $S = \{s_1, s_2, ..., s_N\}$ - states space
- $Z = (Z_1, Z_2, ..., Z_T) \in S^T$ - sequence of states
- $O = \{o_1, o_2, ..., o_M\}$ - observation space
- $X = (X_1, X_2, ..., X_T) \in O^T$ sequence of observations

2. Model assumptions

1. Markov property: $P(Z_t|Z_{t-1}, ..., Z_1) = P(Z_t|Z_{t-1})$
2. Homogeneous Markov chain:
   $P(Z_{t+m} = s_j|Z_{t+m-1} = s_i) =$
   $= P(Z_t = s_j|Z_{t-1} = s_i) = P(Z_1 = s_j|Z_0 = s_i)$
3. Independent observation assumption:
   The observation at time t, i.e. X_t depends only on the state of the Markov chain at time t, i.e. Z_t.

$$P(X_t = o_k|X_1, ..., X_T, Z_1, ..., Z_T) = P(X_t = o_k|Z_t)$$

3. Parameters

- Transition matrix A - $|S| \times |S|$ dimensional probability transition matrix of the hidden Markov chain, i.e. $a_{ij} = P(Z_t = s_j|Z_{t-1} = s_i), \quad i, j = 1, ..., |S|$
- Probability matrix B - $|S|\times|O|$ dimensional matrix describing the probability of the observations, given the state of the hidden Markov chain,
  i.e. $b_{jk} = P(X_t = o_k|Z_t = s_i), \quad j = 1, ..., |S|;$
  $k = 1, ..., |O|$
- Initial distribution π - $|O|$ dimensional probability vector for the initial state
  i.e. $\pi_I = P(Z_0 = s_i), \quad I = 1, ..., |S|$

These 3 parameters above fully describe a HMM, denoted by $\lambda(A, B, \pi)$.

B. HMM as Distribution Forecast

The forecasting problem formulation is following: Given an observation sequence $X_1, ..., X_T$, find the probability distribution of a future observation X_{T+H} for t forecasting horizon H, therefore calculate $P(X_{T+H}|X_1, ..., X_T)$. This problem can be divided in two sub problems. First finding the optimal HMM describing the given observations, thus training, and secondly computing the desired probabilities.

1. Training

Training a HMM is one of the 3 main questions regarding hidden Markov Processes and can be solved by the Baum-Welch Algorithm, as described in Rabiner (1989) an specialized form of an Expectation-Maximization Algorithm. It maximizes the likelihood for a given observation series,

$$\max_{A,B,\pi} P(X_1, ..., X_T|\lambda(A, B, \pi)$$

The Baum-Welch Algorithm utilizes the Markov property and independence of observations to calculate iteratively the optimal HMM parameters. Note, that it is only proven to find a local maxima, in practice however this is sufficient. The computational complexity of the Baum-Welch algorithm is of degree $O(N^2 \times T)$ placing in practice a boundary to state space and length of training data size.

2. Compute Distribution Forecast

Given the observation sequence $X_1, ..., X_T$ and a HMM $\lambda(A, B, \pi)$, the distribution of X_{T+h} is derived analytically,

$$\begin{aligned} P(X_{T+H}|X_1, ..., X_T) &= \\ = P(Z_{T+H}|X_1, ..., X_T)\cdot &= P(X_{T+H}|Z_{T+H}, X_1, ..., X_T) \\ = P(Z_{T+H-1}|X_1, ..., X_T) \cdot &P(Z_{T+H}|Z_{T+H-1}, X_1, ..., X_T) \\ &\cdot P(X_{T+H}|Z_{T+H}) = \\ = P(Z_{T+H-1}|X_1, ..., X_T) \cdot &P(Z_{T+H}|Z_{T+H-1}) \cdot B = \\ = P(Z_{T+H-1}|X_1, ..., X_T) \cdot A \cdot B &= \quad ... \quad = \\ &= P(Z_T|X_1, ..., X_T) \cdot A^H \cdot B \end{aligned}$$

The probabilities of the predicted observation X_{T+H} is deconstructed by the probabilities of the hidden state Z_T at time T multiplied by H times the transition matrix and observation matrix. The conditional probability $P(Z_T|X_1, ..., X_T)$ can be computed with the so-called Forward algorithm, see Rabiner(1989).

III. METHODOLOGY: HMM IN PROBABILISTIC LOAD FORECASTING

A. Model setup and Hyperparameters

- discrete HMM
- Hidden State Space

- unknown, but interpretable: Households specific electric demand actions (Cooking, sleeping,...)
- Possible Decoding: For given HMM and Observation Sequence, the most probable state sequence can be computed (Decoding). With additional information about the household's behavior during the evaluation time, states and behavior can be connected. Thus, HMM models are no "black"-box model and provide insights.
- Hyperparameter: Numbers of States

- Observations:
  - Time series data of load profiles (Watt)
  - discrete
- optional more dimension: Timedata

B. Training

- Input: Timeseries of observation
- Output: HMM $\lambda(A, B, \pi)$
- Training data
  - Avoid seasonal/daytime biases
    - Restricted on full days
    - Include full year or handle specific seasons seperatly
- Computation by Baum-Welch Algorithm
  - For given Observations and fixed number of hidden states, optimizes Log-likelihood of HMM
  - Computational Complexity
  - Algorithm specific Parameters:
    - fixed max Iteration = 100
    - Init Values generated randomly

C. Forecast

- Input: historic Observation sequence $X_1, X_2..., X_T$: HMM from training $\lambda(A, B, \pi)$
- Output: Discrete probability distribution for X_{T+H}
- Computation
  - via Forward algorithm and matrix multiplication as described in Section 2.B.2
  - Init distribution λ of HMM for Test is set to the stationary distribution of the hidden Markov Chain defined by the Eigenvector of A, because the known observation sequence for prediction can differ from the observations used in the training.

D. Evaluation Metrics

The evaluation metrics of probabilistic forecasts are not as straightforward as those applied for point forecasts. While in point forecasting, the distance between the point prediction and observed value is measured, the aim of the distribution forecast is to match the true underlying distribution, which, however, only "materializes" one observation per instance. Hence, the task is to measure the distance between distribution forecast and observation, and with sufficiently many evaluations, derive comparable results. These measures are called scoring rules, see Nowotarski(2018) for a detailed approach. In probabilistic load forecasting, the Continuous Ranked Probability Score (CRPS) has asserted itself in many applications, notebly here is the Global Energy Forecasting Competition (GEFCom2014) summarized in Hong(2016). Its original form is defined by:

$$\mathbf{CRPS}(\hat{F}, y) = \int_{-\infty}^{\infty} \hat{F}(x) - \mathbb{1}_{\{x<y\}} \, dx,$$

where $\hat{F}$ describes the predicted cumulative distribution function and y is the observed value. In this paper, we apply a variation of the CRPS, which can be applied for quantile predictions. This is formulated together with another popular scoring rule, the Pinball score,

$$\mathbf{CRPS}(\hat{F}, y) = \sum_q \mathbf{Pinball}\big(\hat{Q}(q), y, q\big)$$

with $\mathbf{Pinball}\big(\hat{Q}(q), y, q\big)$:

$$= \begin{cases} (1-q) \cdot (\hat{Q}(q) - y), & \text{if } y < \hat{Q}(q) \\ q \cdot (y - \hat{Q}(q)), & \text{if } y \geq \hat{Q}(q) \end{cases}$$

where q denotes a quantile and $\hat{Q}(q)$ the predicted value for q of the distribution prediction. For each evaluation, we calculate the quantiles 0, 0.01, 0.02, ... 0.99, 1 from our predicted distribution and compute the CRPS. To achieve comparable results, multiple evaluations are needed, and then the mean CRPS is taken. Additionally, for hyperparameter tuning, the Log-Likelihood is used to compare the fit of HMM models on training and test datasets. A big difference between training and test data suggests overfitting. The Log-Likelihood $P(O|\lambda)$ of a given HMM λ and an observation sequence can be computed by the Forward Algorithm.

Furthermore, to judge the reliability of the prediction distributions, PIT-histograms (Probability Integral Transforms) are applied. A PIT-Histogram plots the "materialized" quantiles of a set of probability distributions and its corresponding true value into a histogram. For the true distribution, all quantiles appear with a similar frequency, thus all bars of the PIT histogram should show bars with nearly the same height. If the predicted distribution is strongly skewed, like over- or underestimating, the PIT-histogram is not balanced and biases can easily be detected.

IV. USECASE AND MODEL DESIGN

A. Data

- WPuQ Data set: Schlemminger(2022)
  - 38 Households in Germany 2018-2020
  - Includes data: electric power load, heat pump loads, elementary weather and photovoltaic data
- Data usage in paper
  - 15 out of 38 households selected
    - Data fully available from June 1st, 2018 to December 31st, 2020
    - Less than 0.1% of data was interpolated
  - Only regular electric power load was used (no heat pump load)
  - Data resolution in quarter hours
    => 15 timeseries of 90,000 quarter hours of electric power load (Watt)
- Discretization:
  - Data was discretized in a logarithmic manner
  - For data ≤ 2500 Watts:
    - Rounded to the nearest 100 digit (200, 1300, 2400, ..)
  - For data > 2500 Watts:
    - Rounded to the nearest 500 digit (3500, 5000,...)
- Split Training and Test Data:
  - To avoid problems of seasonal biases, only full year data is used for training as well as for testing
    - Training data: December 1st, 2018 - November 30th, 2019 (denoted Year 1)
    - Test data: December 1st, 2019 - November 30th, 2020 (denoted Year 2)
- For season-specific models, training and test data are divided into 4 seasons by months:
  - Spring (March, April, May)
  - Summer (June, July, August)
  - Fall (September, October, November)
  - Winter (December, January, February)
- Split data for hyperparameter tuning and benchmarking:
  - 5 households to evaluate the best hyperparameter setting
  - 10 households for benchmarking
- Daytime information:
  - For models including daytime information, the following setup is proposed:
  - The 96 quarter hours of the day are divided into p periods
    - Example for $p = 2$: 00:00 - 11:45 (1st period), 12:00 - 23:45 (2nd period)
  - Each observation of the timeseries gets its period number as a 2nd dimension
    - i.e., (200 Watts, 2) for an observation at 14:00 with $p = 2$

B. Models

This section discusses the model design for 3 different proposed HMM models: a simple basis model, season-specific models, and daytime model including daytime information. For each model, the hyperparameter settings are investigated for 5 households, and the best performing hyperparameter per model is used for the benchmark tests.

- **Basis model**
  - 1 model for the full year
  - 1-dimensional Observations: power load
  - Training on Year 1; Test on Year 2
  - Hyperparameters: Number of states N
- **Seasonal model**
  - 4 models (1 for each season)
  - 1-dimensional Observations: power load
  - Training on months of the season in Year 1; Test on months of the season in Year 2
  - Hyperparameter: Number of states N
- **Daytime model**
  - 1 model for full year
  - 2-dimensional observations: power load + daytime information
  - Hyperparameters: Number of states N, number of periods in a day P (= time stamps)

C. Evaluation and Benchmarking

- Sliding-Window One-Step-Ahead prediction
  - All tests are conducted with a sliding window one-step-ahead prediction for comparability.
  - Step-ahead prediction:
    * A prediction will be made for one time step (one quarter hour) into the future
    * Forecast Horizon $H = 1$
  - Sliding-window:
    * The historic observations available for each single prediction are the last 7 days
    * Fixed number of historic observations $T = 96 \times 7$
    * REMARK: For computational reasons, the sliding window condition is weakened during the phase of hyperparameter tuning due to the large number of models. For generating the forecast during hyperparameter tuning we consider the whole historical data available until the evaluation date, thus with later evaluation dates the length of the known observation increases. With this method, we can reuse information from previous computations (namely the forward probabilities) and optimize the computational time by a factor of 700.
- Evaluation
  - Hyperparameter selection:
    * For each household, model and hyperparameter setting the log-likelihood for training and test data and the Mean-CRPS are calculated.
      (FOR ZSOLT: In the presented tables I also added the MAE of the expectation point forecast for further discussion)
    * The comparison between log-likelihood of training and test data displays possible overfittings. The Mean-CRPS is used for measuring the general performance of the models forecast.
    * For each model and household the optimal hyperparameter settings are selected by best Mean-CRPS.
  - Benchmarking:
    * Each HMM model is trained with the optimal hyperparameter settings derived from the hyperparameter selection.
    * Comparison by Mean-CRPS
- Benchmarking models
  - See Botman (2025)
    * Similar evaluation framework (CRPS Evaluation, sliding-window prediction)
    * Benchmarking with several state-of-the-art models on 3 different datasets
  - Baseline model
    * Empirical distribution for each quarter hour of the day
      · i.e., the prediction for an observation at 13:45 is made by the empirical distribution of all observations at 13:45 of the training data.
  - State-of-the-art Model
    * Linear quantile regression / Quantile Regression Averaging / Feedforward Neural Network (FFNN) / Bernstein Normalizing Flows (BNF)
    * Not yet chosen

Model	Hyperparameter	LogLike-Train	LogLike-Test	MAE	Mean CRPS
basismodel_hh(1)	states(5)	-41481	-44324	133	4355
basismodel_hh(1)	states(10)	-39066	-42195	120	3983
basismodel_hh(1)	states(15)	-38168	-41614	120	3945
basismodel_hh(1)	states(20)	-37826	-41375	118	3906
basismodel_hh(1)	states(25)	-37373	-41184	119	3925
basismodel_hh(1)	states(30)	-36965	-40900	120	3909
basismodel_hh(1)	states(40)	-36397	-40845	117	3856
basismodel_hh(1)	states(50)	-36757	-41147	118	3876
basismodel_hh(1)	states(60)	-36044	-40673	117	3870

FIG. 1. Basis model evaluation for household Nr.1

V. RESULTS & DISCUSSION

A. HMM Models Hyperparameter Selection

- Example: Figure 1 shows an example evaluation for the basis model of household Nr.1. The optimal number of states is 40 with a Mean-CRPS of 3856, highlighted by yellow. The full table of all households and models is added to the appendix.

1. Basis Model

household	optimal states	Mean-CRPS
1	40	3856
2	40	3977
3	40	9558
4	40	6809
5	60	3119

Optimal states for general basis model: 40

2. Season Model

- Because there is a model for each season, the optimal states consist of 4 numbers representing the optimal number of states for spring, summer, fall and winter.
- The total Mean-CRPS is computed by the mean of the Mean-CRPS of the 4 season. This way, the season model approach can be compared to the other models.

household	optimal states	Mean-CRPS
1	10/20/15/15	3946
2	10/15/15/15	4048
3	10/20/15/15	9894
4	20/10/20/15	7282
5	40/20/40/20	3300

Optimal states for general season model: 10/20/15/15

3. Daytime Model

- NOT FULLY EVALUATED
- By adding the daytime information as a second dimension to the observations, the size of the observation space is multiplied by the number of periods. With increased size also the optimal number of states is greater, which also takes greater computational effort.

household	optimal periods/states	Mean-CRPS
1	10/20	—
2	—	—
3	—	—
4	—	—
5	—	—

B. Benchmarks

No precise benchmarks have been conducted yet. Preliminary experiments show an improvement to the baseline model of around 10-20%. Although in the literature state-of-the-art models achieve an improvement of 30% with a forecasting horizon of one day.

VI. SUMMARY AND OUTLOOK

- This paper constructs a method to utilize HMM in probabilistic forecasting.
- As a preliminary result so far, it can be said that HMM models applied on household electric power load tend to over fit with a growing number of states, indicating an optimal number of states beneath 100. With this (vague) information and considering the complexity and variability of households electricity demand profiles, the limitations of HMM in load forecasting are shown.
- Nonetheless, the experiments and model evaluation have to be completed before making early conclusions. Especially models with lower number of states should be studied to finish this work.

VII. APPENDIX

Model	Hyperparameter	LogLike-Train	LogLike-Test	MAE	Mean CRPS
basismodel_hh(1)	states(5)	-41481	-44324	133	4355
basismodel_hh(1)	states(10)	-39066	-42195	120	3983
basismodel_hh(1)	states(15)	-38168	-41614	120	3945
basismodel_hh(1)	states(20)	-37826	-41375	118	3906
basismodel_hh(1)	states(25)	-37373	-41184	119	3925
basismodel_hh(1)	states(30)	-36965	-40900	120	3909
basismodel_hh(1)	states(40)	-36397	-40845	117	3856
basismodel_hh(1)	states(50)	-36757	-41147	118	3876
basismodel_hh(1)	states(60)	-36044	-40673	117	3870
daytimemodel_hh(1)	ts(4)_states(5)	-52030	-56011	147	4832
daytimemodel_hh(1)	ts(4)_states(10)	-47264	-50806	128	4290
daytimemodel_hh(1)	ts(4)_states(15)	-46062	-49828	126	4123
daytimemodel_hh(1)	ts(4)_states(20)	-45168	-48937	122	4020
daytimemodel_hh(1)	ts(4)_states(25)	-44978	-49028	124	4064
daytimemodel_hh(1)	ts(4)_states(30)	-44330	-48277	122	3987
daytimemodel_hh(1)	ts(4)_states(40)	-43840	-48147	122	3993
daytimemodel_hh(1)	ts(4)_states(50)	-42852	-47718	120	3940
daytimemodel_hh(1)	ts(4)_states(60)	-42778	-48050	121	3975
daytimemodel_hh(1)	ts(4)_states(80)	-42231	-48222	119	3928
daytimemodel_hh(1)	ts(4)_states(100)	-41331	-47740	118	3919
daytimemodel_hh(1)	ts(8)_states(5)	-82970	-87195	148	5144
daytimemodel_hh(1)	ts(8)_states(10)	-55815	-59999	150	4958
daytimemodel_hh(1)	ts(8)_states(15)	-59603	-63973	141	4742
daytimemodel_hh(1)	ts(8)_states(20)	-53239	-57481	143	4633
daytimemodel_hh(1)	ts(8)_states(25)	-50539	-55139	132	4276
daytimemodel_hh(1)	ts(8)_states(30)	-50516	-55260	129	4280
daytimemodel_hh(1)	ts(8)_states(40)	-48723	-53652	130	4169
daytimemodel_hh(1)	ts(8)_states(50)	-48279	-53520	127	4145
daytimemodel_hh(1)	ts(8)_states(60)	-47607	-53446	127	4116
daytimemodel_hh(1)	ts(8)_states(80)	-46351	-52786	122	3996
daytimemodel_hh(1)	ts(8)_states(100)	45969	53608	122	4006
daytimemodel_hh(1)	ts(12)_states(5)	-97501	-103429	145	5183
daytimemodel_hh(1)	ts(12)_states(10)	-73160	-78844	150	4931
daytimemodel_hh(1)	ts(12)_states(15)	-58712	-64358	150	4899
daytimemodel_hh(1)	ts(12)_states(20)	-59543	-65464	145	4656
daytimemodel_hh(1)	ts(12)_states(25)	-55505	-61267	141	4616
daytimemodel_hh(1)	ts(12)_states(30)	-58206	-64760	135	4442
daytimemodel_hh(1)	ts(12)_states(40)	-53096	-59253	137	4393
daytimemodel_hh(1)	ts(12)_states(50)	-52565	-59434	129	4245
daytimemodel_hh(1)	ts(12)_states(60)	-51428	-58575	127	4162
daytimemodel_hh(1)	ts(12)_states(80)	-50147	-58312	125	4084
daytimemodel_hh(1)	ts(12)_states(100)	-48861	-58649	123	4078
daytimemodel_hh(1)	ts(24)_states(5)	-125740	-133572	148	5235
daytimemodel_hh(1)	ts(24)_states(10)	-96549	-104081	148	4978
daytimemodel_hh(1)	ts(24)_states(15)	-85517	-93685	152	5108
daytimemodel_hh(1)	ts(24)_states(20)	-77710	-85315	149	4981
daytimemodel_hh(1)	ts(24)_states(25)	-69101	-77156	148	4885
daytimemodel_hh(1)	ts(24)_states(30)	-66868	-76103	150	4918
daytimemodel_hh(1)	ts(24)_states(40)	-65912	-76223	143	4690
daytimemodel_hh(1)	ts(24)_states(50)	-60609	-72497	139	4647
daytimemodel_hh(1)	ts(24)_states(60)	-59474	-70367	144	4669
daytimemodel_hh(1)	ts(24)_states(80)	-56574	-67811	140	4542
daytimemodel_hh(1)	ts(24)_states(100)	-54965	-69351	138	4480
daytimemodel_hh(1)	ts(96)_states(5)	-163329	-182927	150	5259
daytimemodel_hh(1)	ts(96)_states(10)	-149067	-170255	147	5191
daytimemodel_hh(1)	ts(96)_states(15)	-127411	-148296	149	5245
daytimemodel_hh(1)	ts(96)_states(20)	-125626	-146928	148	5211
daytimemodel_hh(1)	ts(96)_states(25)	-114249	-135340	149	5236
daytimemodel_hh(1)	ts(96)_states(30)	-112005	-134481	150	5252
daytimemodel_hh(1)	ts(96)_states(40)	-93182	-114262	150	5183
daytimemodel_hh(1)	ts(96)_states(50)	-90397	-112153	150	5220
daytimemodel_hh(1)	ts(96)_states(60)	-78976	-100320	148	5118
daytimemodel_hh(1)	ts(96)_states(80)	-69132	-90900	149	5033
daytimemodel_hh(1)	ts(96)_states(100)	-61758	-84286	150	5055
seasonmodel_hh(1)	spring_states(5)	-10442	-11169	138	4478
seasonmodel_hh(1)	spring_states(10)	-9199	-10326	123	4086
seasonmodel_hh(1)	spring_states(15)	-8931	-10339	125	4134
seasonmodel_hh(1)	spring_states(20)	-8788	-10622	125	4148
seasonmodel_hh(1)	spring_states(25)	-8778	-10569	129	4197
seasonmodel_hh(1)	spring_states(30)	-8601	-10751	124	4174
seasonmodel_hh(1)	spring_states(40)	-8442	-10781	126	4199
seasonmodel_hh(1)	spring_states(50)	-8305	-11094	126	4214
seasonmodel_hh(1)	spring_states(60)	-8161	-11146	127	4271
seasonmodel_hh(1)	summer_states(5)	-10241	-11183	114	3907
seasonmodel_hh(1)	summer_states(10)	-9860	-10829	114	3837
seasonmodel_hh(1)	summer_states(15)	-9646	-10786	113	3795
seasonmodel_hh(1)	summer_states(20)	-9539	-10847	112	3746
seasonmodel_hh(1)	summer_states(25)	-9612	-10827	115	3803
seasonmodel_hh(1)	summer_states(30)	-9459	-10817	115	3819
seasonmodel_hh(1)	summer_states(40)	-9066	-10899	121	3862
seasonmodel_hh(1)	summer_states(50)	-8863	-11069	116	3832
seasonmodel_hh(1)	summer_states(60)	-8890	-10973	118	3836
seasonmodel_hh(1)	fall_states(5)	-10181	-10804	124	4044
seasonmodel_hh(1)	fall_states(10)	-9333	-10235	117	3787
seasonmodel_hh(1)	fall_states(15)	-9097	-10180	112	3685
seasonmodel_hh(1)	fall_states(20)	-8833	-10150	113	3720
seasonmodel_hh(1)	fall_states(25)	-8864	-10242	116	3768
seasonmodel_hh(1)	fall_states(30)	-8863	-10202	115	3762
seasonmodel_hh(1)	fall_states(40)	-8435	-10328	114	3721
seasonmodel_hh(1)	fall_states(50)	-8345	-10651	117	3804
seasonmodel_hh(1)	fall_states(60)	-8198	-10765	115	3807
seasonmodel_hh(1)	winter_states(5)	-9647	-10716	142	4746
seasonmodel_hh(1)	winter_states(10)	-8864	-10285	129	4312
seasonmodel_hh(1)	winter_states(15)	-8594	-10214	127	4270
seasonmodel_hh(1)	winter_states(20)	-8433	-10354	131	4346
seasonmodel_hh(1)	winter_states(25)	8400	10345	129	4296
seasonmodel_hh(1)	winter_states(30)	-8265	-10355	134	4390
seasonmodel_hh(1)	winter_states(40)	-8027	-10810	130	4472
seasonmodel_hh(1)	winter_states(50)	-7993	-10767	128	4386
seasonmodel_hh(1)	winter_states(60)	-7875	-11078	131	4413

FIG. 2. Model evaluations for household Nr.1

Model	Hyperparameter	LogLike-Train	LogLike-Test	MAE	Mean CRPS
basismodel_hh(2)	states(5)	-46470	-51888	130	4624
basismodel_hh(2)	states(10)	-41214	-48460	122	4211
basismodel_hh(2)	states(15)	-40270	-47541	116	4002
basismodel_hh(2)	states(20)	-40088	-47360	118	4035
basismodel_hh(2)	states(25)	-40153	-47073	118	4023
basismodel_hh(2)	states(30)	-39945	-47237	118	4008
basismodel_hh(2)	states(40)	-39739	-46890	116	3977
basismodel_hh(2)	states(50)	-39622	-47336	116	3990
basismodel_hh(2)	states(60)	-39560	-47338	116	3981
daytimemodel_hh(2)	ts(4)_states(5)	-65535	-67940	151	5645
daytimemodel_hh(2)	ts(4)_states(10)	-53272	-59907	138	4790
daytimemodel_hh(2)	ts(4)_states(15)	-51002	-58351	130	4638
daytimemodel_hh(2)	ts(4)_states(20)	-48621	-56951	134	4612
daytimemodel_hh(2)	ts(4)_states(25)	-47191	-55600	125	4352
daytimemodel_hh(2)	ts(4)_states(30)	-47344	-55791	123	4298
daytimemodel_hh(2)	ts(4)_states(40)	-45198	-54233	119	4110
daytimemodel_hh(2)	ts(4)_states(50)	-44879	-54782	118	4127
daytimemodel_hh(2)	ts(4)_states(60)	-44566	-54543	118	4138
daytimemodel_hh(2)	ts(8)_states(5)	-96384	-99298	157	5796
daytimemodel_hh(2)	ts(8)_states(10)	-65060	-69387	152	5385
daytimemodel_hh(2)	ts(8)_states(15)	-63548	-67662	145	5235
daytimemodel_hh(2)	ts(8)_states(20)	-59575	-64984	138	4734
daytimemodel_hh(2)	ts(8)_states(25)	-55344	-63363	135	4667
daytimemodel_hh(2)	ts(8)_states(30)	-55328	-62503	131	4604
daytimemodel_hh(2)	ts(8)_states(40)	-51027	-60440	124	4338
daytimemodel_hh(2)	ts(8)_states(50)	-50786	-60437	122	4285
daytimemodel_hh(2)	ts(8)_states(60)	-50209	-59612	119	4158
daytimemodel_hh(2)	ts(12)_states(5)	-101948	-105518	151	5560
daytimemodel_hh(2)	ts(12)_states(10)	-80358	-85022	152	5538
daytimemodel_hh(2)	ts(12)_states(15)	-69070	-73140	150	5440
daytimemodel_hh(2)	ts(12)_states(20)	-65754	-71915	146	5224
daytimemodel_hh(2)	ts(12)_states(25)	-65729	-72724	142	4945
daytimemodel_hh(2)	ts(12)_states(30)	-59637	-67053	133	4683
daytimemodel_hh(2)	ts(12)_states(40)	-56605	-66233	130	4590
daytimemodel_hh(2)	ts(12)_states(50)	-56674	-66514	135	4676
daytimemodel_hh(2)	ts(12)_states(60)	-53672	-64806	128	4470
daytimemodel_hh(2)	ts(24)_states(5)	-129370	-135609	152	5754
daytimemodel_hh(2)	ts(24)_states(10)	-101400	-106920	154	5536
daytimemodel_hh(2)	ts(24)_states(15)	-93897	-99475	151	5623
daytimemodel_hh(2)	ts(24)_states(20)	-88637	-94994	151	5580
daytimemodel_hh(2)	ts(24)_states(25)	-80248	-86762	148	5424
daytimemodel_hh(2)	ts(24)_states(30)	-75708	-82160	149	5387
daytimemodel_hh(2)	ts(24)_states(40)	-72936	-80241	143	5144
daytimemodel_hh(2)	ts(24)_states(50)	-70193	-81384	139	4970
daytimemodel_hh(2)	ts(24)_states(60)	-65721	-77605	139	4935
daytimemodel_hh(2)	ts(96)_states(5)	-176526	-195653	160	5919
daytimemodel_hh(2)	ts(96)_states(10)	-154345	-174428	158	5902
daytimemodel_hh(2)	ts(96)_states(15)	-137579	-158058	156	5841
daytimemodel_hh(2)	ts(96)_states(20)	-133867	-154651	159	5907
daytimemodel_hh(2)	ts(96)_states(25)	-123803	-143727	153	5757
daytimemodel_hh(2)	ts(96)_states(30)	-117468	-138955	153	5786
daytimemodel_hh(2)	ts(96)_states(40)	-102672	-122000	158	5831
daytimemodel_hh(2)	ts(96)_states(50)	-96656	-118598	153	5691
daytimemodel_hh(2)	ts(96)_states(60)	-91797	-112561	154	5795
seasonmodel_hh(2)	spring_states(5)	-11765	-12015	128	4304
seasonmodel_hh(2)	spring_states(10)	-11158	-11483	115	3937
seasonmodel_hh(2)	spring_states(15)	-11027	-11489	115	3940
seasonmodel_hh(2)	spring_states(20)	-11030	-11478	116	3959
seasonmodel_hh(2)	spring_states(25)	-10937	-11516	115	3939
seasonmodel_hh(2)	spring_states(30)	-10968	-11524	116	3965
seasonmodel_hh(2)	spring_states(40)	-10747	-11665	116	3984
seasonmodel_hh(2)	spring_states(50)	-10668	-11794	116	3991
seasonmodel_hh(2)	spring_states(60)	-10575	-11898	117	4013
seasonmodel_hh(2)	summer_states(5)	-9372	-14415	150	5267
seasonmodel_hh(2)	summer_states(10)	-8694	-12942	133	4584
seasonmodel_hh(2)	summer_states(15)	-8365	-12618	125	4287
seasonmodel_hh(2)	summer_states(20)	-8374	-12641	129	4319
seasonmodel_hh(2)	summer_states(25)	-8364	-12710	127	4290
seasonmodel_hh(2)	summer_states(30)	-8257	-12914	129	4396
seasonmodel_hh(2)	summer_states(40)	-8132	-12988	126	4316
seasonmodel_hh(2)	summer_states(50)	-7966	-13373	130	4451
seasonmodel_hh(2)	summer_states(60)	-7955	-13376	128	4436
seasonmodel_hh(2)	fall_states(5)	-10649	-13915	148	5220
seasonmodel_hh(2)	fall_states(10)	-10071	-13359	139	4670
seasonmodel_hh(2)	fall_states(15)	-9784	-13231	128	4474
seasonmodel_hh(2)	fall_states(20)	-9759	-13251	133	4523
seasonmodel_hh(2)	fall_states(25)	-9705	-13368	133	4561
seasonmodel_hh(2)	fall_states(30)	-9668	-13293	132	4535
seasonmodel_hh(2)	fall_states(40)	-9444	-13419	132	4539
seasonmodel_hh(2)	fall_states(50)	-9375	-13749	135	4665
seasonmodel_hh(2)	fall_states(60)	-9277	-14056	132	4635
seasonmodel_hh(2)	winter_states(5)	-11589	-11082	122	4032
seasonmodel_hh(2)	winter_states(10)	-10696	-10426	101	3565
seasonmodel_hh(2)	winter_states(15)	-10613	-10401	101	3498
seasonmodel_hh(2)	winter_states(20)	-10600	-10447	102	3590
seasonmodel_hh(2)	winter_states(25)	-10358	-10585	99	3482
seasonmodel_hh(2)	winter_states(30)	-10388	-10393	102	3494
seasonmodel_hh(2)	winter_states(40)	-10062	-10956	105	3614
seasonmodel_hh(2)	winter_states(50)	-9906	-11061	104	3578
seasonmodel_hh(2)	winter_states(60)	-9792	-11083	102	3562

FIG. 3. Model evaluations for household Nr.2

Model	Hyperparameter	LogLike-Train	LogLike-Test	MAE	Mean CRPS
basismodel_hh(3)	states(5)	-65174	-68025	321	10547
basismodel_hh(3)	states(10)	-63814	-66352	304	10082
basismodel_hh(3)	states(15)	-62784	-65645	289	9713
basismodel_hh(3)	states(20)	-62486	-65394	288	9693
basismodel_hh(3)	states(25)	-62260	-65395	287	9682
basismodel_hh(3)	states(30)	-62054	-65263	286	9604
basismodel_hh(3)	states(40)	-61156	-64885	284	9558
basismodel_hh(3)	states(50)	-61042	-65023	284	9565
basismodel_hh(3)	states(60)	-60690	-65139	284	9608
daytimemodel_hh(3)	ts(4)_states(5)	-92144	-96763	400	13175
daytimemodel_hh(3)	ts(4)_states(10)	-74726	-78890	367	11818
daytimemodel_hh(3)	ts(4)_states(15)	-71706	-75620	348	11010
daytimemodel_hh(3)	ts(4)_states(20)	-70457	-74678	333	10578
daytimemodel_hh(3)	ts(4)_states(25)	-69488	-73532	322	10386
daytimemodel_hh(3)	ts(4)_states(30)	-68756	-72975	309	10065
daytimemodel_hh(3)	ts(4)_states(40)	-67698	-72000	295	9840
daytimemodel_hh(3)	ts(4)_states(50)	-67638	-72137	299	9889
daytimemodel_hh(3)	ts(4)_states(60)	-68274	-73569	299	10000
daytimemodel_hh(3)	ts(8)_states(5)	-106762	-111494	392	13123
daytimemodel_hh(3)	ts(8)_states(10)	-86945	-91502	382	12560
daytimemodel_hh(3)	ts(8)_states(15)	-80928	-85526	365	12020
daytimemodel_hh(3)	ts(8)_states(20)	-79162	-83638	352	11501
daytimemodel_hh(3)	ts(8)_states(25)	-76992	-81604	352	11483
daytimemodel_hh(3)	ts(8)_states(30)	-75892	-81184	341	10994
daytimemodel_hh(3)	ts(8)_states(40)	-73554	-78566	319	10442
daytimemodel_hh(3)	ts(8)_states(50)	-72189	-77713	308	10092
daytimemodel_hh(3)	ts(8)_states(60)	-72812	-78858	320	10369
daytimemodel_hh(3)	ts(12)_states(5)	-122733	-127883	406	13520
daytimemodel_hh(3)	ts(12)_states(10)	-97133	-102018	385	12966
daytimemodel_hh(3)	ts(12)_states(15)	-89426	-94873	376	12580
daytimemodel_hh(3)	ts(12)_states(20)	-83827	-89152	362	11819
daytimemodel_hh(3)	ts(12)_states(25)	-82463	-87650	362	11622
daytimemodel_hh(3)	ts(12)_states(30)	-80755	-86365	349	11256
daytimemodel_hh(3)	ts(12)_states(40)	-79280	-84563	341	11063
daytimemodel_hh(3)	ts(12)_states(50)	-77244	-84205	331	10741
daytimemodel_hh(3)	ts(12)_states(60)	-77213	-83762	332	10849
daytimemodel_hh(3)	ts(24)_states(5)	-152126	-159426	405	13539
daytimemodel_hh(3)	ts(24)_states(10)	-123297	-129846	392	13128
daytimemodel_hh(3)	ts(24)_states(15)	-112174	-118387	390	12959
daytimemodel_hh(3)	ts(24)_states(20)	-105122	-112701	381	12751
daytimemodel_hh(3)	ts(24)_states(25)	-98446	-105762	377	12600
daytimemodel_hh(3)	ts(24)_states(30)	-95193	-102976	375	12457
daytimemodel_hh(3)	ts(24)_states(40)	-90614	-100098	364	12057
daytimemodel_hh(3)	ts(24)_states(50)	-87783	-96548	360	11747
daytimemodel_hh(3)	ts(24)_states(60)	-86714	-96231	357	11663
daytimemodel_hh(3)	*ts(96)_states(5)*	*-230112*	*-245973*	*408*	*13688*
daytimemodel_hh(3)	*ts(96)_states(10)*	*-230087*	*-245879*	*408*	*13688*
daytimemodel_hh(3)	*ts(96)_states(15)*	*-230196*	*-245977*	*408*	*13688*
daytimemodel_hh(3)	*ts(96)_states(20)*	*-230629*	*-246302*	*408*	*13688*
daytimemodel_hh(3)	*ts(96)_states(25)*	*-230792*	*-246427*	*408*	*13687*
daytimemodel_hh(3)	*ts(96)_states(30)*	*-230906*	*-246528*	*408*	*13688*
daytimemodel_hh(3)	*ts(96)_states(40)*	*-230960*	*-246550*	*408*	*13688*
daytimemodel_hh(3)	*ts(96)_states(50)*	*-231088*	*-246677*	*408*	*13688*
daytimemodel_hh(3)	*ts(96)_states(60)*	*-231136*	*-246698*	*408*	*13688*
seasonmodel_hh(3)	spring_states(5)	-17110	-18121	348	11522
seasonmodel_hh(3)	spring_states(10)	-16523	-17621	332	10986
seasonmodel_hh(3)	spring_states(15)	-16255	-17511	315	10712
seasonmodel_hh(3)	spring_states(20)	-16115	-17604	317	10797
seasonmodel_hh(3)	spring_states(25)	-16014	-17760	320	10877
seasonmodel_hh(3)	spring_states(30)	-15856	-17800	328	11034
seasonmodel_hh(3)	spring_states(40)	-15571	-18067	321	10929
seasonmodel_hh(3)	spring_states(50)	-15423	-18226	331	11258
seasonmodel_hh(3)	spring_states(60)	-15240	-18537	326	11191
seasonmodel_hh(3)	summer_states(5)	-14445	-16502	300	9725
seasonmodel_hh(3)	summer_states(10)	-13693	-16162	279	9256
seasonmodel_hh(3)	summer_states(15)	-13465	-16336	287	9354
seasonmodel_hh(3)	summer_states(20)	-13299	-16452	285	9369
seasonmodel_hh(3)	summer_states(25)	-13180	-16196	273	9092
seasonmodel_hh(3)	summer_states(30)	-12970	-16464	279	9290
seasonmodel_hh(3)	summer_states(40)	-12766	-16656	285	9366
seasonmodel_hh(3)	summer_states(50)	-12663	-16906	285	9435
seasonmodel_hh(3)	summer_states(60)	-12539	-17017	287	9549
seasonmodel_hh(3)	fall_states(5)	-16487	-17024	324	10542
seasonmodel_hh(3)	fall_states(10)	-16129	-16780	305	10122
seasonmodel_hh(3)	fall_states(15)	-15828	-16627	292	9906
seasonmodel_hh(3)	fall_states(20)	-15724	-16738	294	9966
seasonmodel_hh(3)	fall_states(25)	-15665	-16740	293	9963
seasonmodel_hh(3)	fall_states(30)	-15409	-16893	295	10027
seasonmodel_hh(3)	fall_states(40)	-15224	-17030	297	10113
seasonmodel_hh(3)	fall_states(50)	-15018	-17108	298	10136
seasonmodel_hh(3)	fall_states(60)	-14813	-17395	302	10330
seasonmodel_hh(3)	winter_states(5)	-16876	-17009	323	10552
seasonmodel_hh(3)	winter_states(10)	-16379	-16676	311	10182
seasonmodel_hh(3)	winter_states(15)	-16080	-16635	293	9871
seasonmodel_hh(3)	winter_states(20)	-15901	-16654	294	9867
seasonmodel_hh(3)	winter_states(25)	-15798	-16827	300	9957
seasonmodel_hh(3)	winter_states(30)	-15649	-16989	301	10023
seasonmodel_hh(3)	winter_states(40)	-15403	-17365	303	10163
seasonmodel_hh(3)	winter_states(50)	-15180	-17299	299	10174
seasonmodel_hh(3)	winter_states(60)	-15046	-17576	309	10356

FIG. 4. Model evaluations for household Nr.3

Model	Hyperparameter	LogLike-Train	LogLike-Test	MAE	Mean CRPS
basismodel_hh(4)	states(5)	-51976	-59854	256	8024
basismodel_hh(4)	states(10)	-49515	-58093	216	7225
basismodel_hh(4)	states(15)	-49604	-57414	214	7087
basismodel_hh(4)	states(20)	-48498	-55366	222	7082
basismodel_hh(4)	states(25)	-48097	-54763	203	6824
basismodel_hh(4)	states(30)	-48127	-54630	207	6862
basismodel_hh(4)	states(40)	-47716	-54141	208	6809
basismodel_hh(4)	states(50)	-47968	-55918	218	7054
basismodel_hh(4)	states(60)	-47347	-54333	206	6850
daytimemodel_hh(4)	ts(4)_states(5)	-73466	-84843	273	9643
daytimemodel_hh(4)	ts(4)_states(10)	-60717	-70933	259	8888
daytimemodel_hh(4)	ts(4)_states(20)	-55315	-65497	234	7494
daytimemodel_hh(4)	ts(4)_states(15)	-57291	-67993	253	8282
daytimemodel_hh(4)	ts(4)_states(25)	-54602	-64928	224	7347
daytimemodel_hh(4)	ts(4)_states(30)	-54197	-64083	217	7174
daytimemodel_hh(4)	ts(4)_states(40)	-53575	-64089	225	7258
daytimemodel_hh(4)	ts(4)_states(50)	-53196	-65279	221	7235
daytimemodel_hh(4)	ts(4)_states(60)	-52527	-64132	220	7196
daytimemodel_hh(4)	ts(8)_states(5)	-89541	-102624	258	9584
daytimemodel_hh(4)	ts(8)_states(10)	-71616	-84404	268	9350
daytimemodel_hh(4)	ts(8)_states(15)	-64647	-78561	266	9354
daytimemodel_hh(4)	ts(8)_states(20)	-61722	-75066	246	8292
daytimemodel_hh(4)	ts(8)_states(25)	-60479	-72840	251	8073
daytimemodel_hh(4)	ts(8)_states(30)	-59851	-73511	247	7961
daytimemodel_hh(4)	ts(8)_states(40)	-58298	-71850	246	7797
daytimemodel_hh(4)	ts(8)_states(50)	-57577	-71982	233	7556
daytimemodel_hh(4)	ts(8)_states(60)	-57798	-72876	233	7594
daytimemodel_hh(4)	ts(12)_states(5)	-99226	-112486	265	9634
daytimemodel_hh(4)	ts(12)_states(10)	-86753	-100002	262	9550
daytimemodel_hh(4)	ts(12)_states(15)	-73291	-87177	264	9362
daytimemodel_hh(4)	ts(12)_states(20)	-67340	-81087	258	8607
daytimemodel_hh(4)	ts(12)_states(25)	-69037	-84973	264	8873
daytimemodel_hh(4)	ts(12)_states(30)	-64329	-79293	263	8675
daytimemodel_hh(4)	ts(12)_states(40)	-63156	-79799	253	8498
daytimemodel_hh(4)	ts(12)_states(50)	-61363	-77155	249	7933
daytimemodel_hh(4)	ts(12)_states(60)	-60755	-77774	240	7801
daytimemodel_hh(4)	ts(24)_states(5)	-133886	-152823	276	10180
daytimemodel_hh(4)	ts(24)_states(10)	-108117	-126868	262	9543
daytimemodel_hh(4)	ts(24)_states(15)	-98911	-118452	265	9689
daytimemodel_hh(4)	ts(24)_states(20)	-88081	-108252	263	9529
daytimemodel_hh(4)	ts(24)_states(25)	-83457	-103530	262	9316
daytimemodel_hh(4)	ts(24)_states(30)	-76958	-98210	264	9224
daytimemodel_hh(4)	ts(24)_states(40)	-73981	-94211	265	9223
daytimemodel_hh(4)	ts(24)_states(50)	-70594	-92496	259	8757
daytimemodel_hh(4)	ts(24)_states(60)	-68402	-91090	258	8631
daytimemodel_hh(4)	ts(96)_states(5)	-174194	-208739	276	10186
daytimemodel_hh(4)	ts(96)_states(10)	-156991	-192140	273	10102
daytimemodel_hh(4)	ts(96)_states(15)	-146278	-184223	275	10143
daytimemodel_hh(4)	ts(96)_states(20)	-134374	-170894	275	10164
daytimemodel_hh(4)	ts(96)_states(25)	-123384	-159044	273	10024
daytimemodel_hh(4)	ts(96)_states(30)	-121759	-159358	274	10035
daytimemodel_hh(4)	ts(96)_states(40)	-102767	-138707	270	9978
daytimemodel_hh(4)	ts(96)_states(50)	-96573	-133556	272	9998
daytimemodel_hh(4)	ts(96)_states(60)	-87942	-123934	272	9989
seasonmodel_hh(4)	spring_states(5)	-12595	-15483	252	8307
seasonmodel_hh(4)	spring_states(10)	-11746	-14960	235	7861
seasonmodel_hh(4)	spring_states(15)	-11792	-15029	229	7802
seasonmodel_hh(4)	spring_states(20)	-11293	-14880	224	7690
seasonmodel_hh(4)	spring_states(25)	-11134	-15113	232	7839
seasonmodel_hh(4)	spring_states(30)	-11045	-14999	225	7763
seasonmodel_hh(4)	spring_states(40)	-11023	-15346	233	7920
seasonmodel_hh(4)	spring_states(50)	-10849	-15545	239	8044
seasonmodel_hh(4)	spring_states(60)	-10608	-16072	240	8207
seasonmodel_hh(4)	summer_states(5)	-12815	-17392	273	8214
seasonmodel_hh(4)	summer_states(10)	-12450	-17189	213	7561
seasonmodel_hh(4)	summer_states(15)	-12326	-17039	218	7583
seasonmodel_hh(4)	summer_states(20)	-12251	-17375	215	7626
seasonmodel_hh(4)	summer_states(25)	-12123	-17232	220	7886
seasonmodel_hh(4)	summer_states(30)	-12053	-17771	207	7679
seasonmodel_hh(4)	summer_states(40)	-11904	-17435	216	7827
seasonmodel_hh(4)	summer_states(50)	-11825	-17901	242	8044
seasonmodel_hh(4)	summer_states(60)	-11552	-16503	269	8057
seasonmodel_hh(4)	fall_states(5)	-13021	-14604	245	7811
seasonmodel_hh(4)	fall_states(10)	-12099	-13851	215	7061
seasonmodel_hh(4)	fall_states(15)	-11936	-13732	213	7035
seasonmodel_hh(4)	fall_states(20)	-11855	-13584	216	7033
seasonmodel_hh(4)	fall_states(25)	-11635	-13836	218	7074
seasonmodel_hh(4)	fall_states(30)	-11708	-13803	217	7142
seasonmodel_hh(4)	fall_states(40)	-11485	-13789	216	7034
seasonmodel_hh(4)	fall_states(50)	-11454	-13835	216	7087
seasonmodel_hh(4)	fall_states(60)	-11042	-14340	216	7244
seasonmodel_hh(4)	winter_states(5)	-12792	-13326	234	7679
seasonmodel_hh(4)	winter_states(10)	-12136	-12843	223	7262
seasonmodel_hh(4)	winter_states(15)	-11947	-12691	207	6846
seasonmodel_hh(4)	winter_states(20)	-11873	-12719	211	6969
seasonmodel_hh(4)	winter_states(25)	-11726	-12846	209	6962
seasonmodel_hh(4)	winter_states(30)	-11653	-12871	210	6977
seasonmodel_hh(4)	winter_states(40)	-11561	-12866	209	6971
seasonmodel_hh(4)	winter_states(50)	-11490	-13005	213	7048
seasonmodel_hh(4)	winter_states(60)	-11158	-13550	209	7127

FIG. 5. Model evaluations for household Nr.4

Model	Hyperparameter	LogLike-Train	LogLike-Test	MAE	Mean CRPS
basismodel_hh(5)	states(5)	-40831	-41320	107	3711
basismodel_hh(5)	states(10)	-35683	-37723	100	3338
basismodel_hh(5)	states(15)	-34568	-36811	98	3265
basismodel_hh(5)	states(20)	-34132	-36439	98	3272
basismodel_hh(5)	states(25)	-33521	-35614	95	3169
basismodel_hh(5)	states(30)	-33390	-36027	94	3158
basismodel_hh(5)	states(40)	-32850	-35351	95	3172
basismodel_hh(5)	states(50)	-32578	-35507	95	3176
basismodel_hh(5)	states(60)	-32262	-35335	93	3119
daytimemodel_hh(5)	ts(4)_states(5)	-78349	-75005	180	5722
daytimemodel_hh(5)	ts(4)_states(10)	-51152	-51129	130	4095
daytimemodel_hh(5)	ts(4)_states(15)	-50407	-51643	127	4272
daytimemodel_hh(5)	ts(4)_states(20)	-44082	-45615	103	3499
daytimemodel_hh(5)	ts(4)_states(25)	-42745	-44845	104	3492
daytimemodel_hh(5)	ts(4)_states(30)	-41859	-43699	97	3292
daytimemodel_hh(5)	ts(4)_states(40)	-40717	-43627	99	3313
daytimemodel_hh(5)	ts(4)_states(50)	-39180	-42557	98	3290
daytimemodel_hh(5)	ts(4)_states(60)	-38864	-42626	98	3253
daytimemodel_hh(5)	ts(8)_states(5)	-90119	-86587	176	5648
daytimemodel_hh(5)	ts(8)_states(10)	-71999	-70373	170	5433
daytimemodel_hh(5)	ts(8)_states(15)	-63161	-62458	129	4433
daytimemodel_hh(5)	ts(8)_states(20)	-57590	-57940	127	4261
daytimemodel_hh(5)	ts(8)_states(25)	-52297	-53596	117	3826
daytimemodel_hh(5)	ts(8)_states(30)	-52706	-54306	129	4069
daytimemodel_hh(5)	ts(8)_states(40)	-47569	-49570	101	3441
daytimemodel_hh(5)	ts(8)_states(50)	-45765	-48448	102	3469
daytimemodel_hh(5)	ts(8)_states(60)	-44790	-48350	98	3322
daytimemodel_hh(5)	ts(12)_states(5)	-113083	-110527	177	5473
daytimemodel_hh(5)	ts(12)_states(10)	-89133	-86803	159	5116
daytimemodel_hh(5)	ts(12)_states(15)	-81259	-79321	171	5376
daytimemodel_hh(5)	ts(12)_states(20)	-69794	-69335	145	4617
daytimemodel_hh(5)	ts(12)_states(25)	-62523	-63127	132	4307
daytimemodel_hh(5)	ts(12)_states(30)	-59892	-59985	129	4100
daytimemodel_hh(5)	ts(12)_states(40)	-55655	-56934	114	3687
daytimemodel_hh(5)	ts(12)_states(50)	-51956	-54399	105	3548
daytimemodel_hh(5)	ts(12)_states(60)	-51349	-54794	102	3471
daytimemodel_hh(5)	ts(24)_states(5)	-140162	-140119	189	6147
daytimemodel_hh(5)	ts(24)_states(10)	-114581	-114135	166	5309
daytimemodel_hh(5)	ts(24)_states(15)	-104724	-105615	179	5652
daytimemodel_hh(5)	ts(24)_states(20)	-97022	-97661	175	5478
daytimemodel_hh(5)	ts(24)_states(25)	-90033	-91478	155	4993
daytimemodel_hh(5)	ts(24)_states(30)	-84283	-85210	142	4789
daytimemodel_hh(5)	ts(24)_states(40)	-74192	-75658	141	4345
daytimemodel_hh(5)	ts(24)_states(50)	-71873	-76886	124	4036
daytimemodel_hh(5)	ts(24)_states(60)	-65679	-70330	118	3858
daytimemodel_hh(5)	ts(96)_states(5)	-184436	-192129	202	6532
daytimemodel_hh(5)	ts(96)_states(10)	-166779	-176624	198	6350
daytimemodel_hh(5)	ts(96)_states(15)	-153312	-164393	205	6644
daytimemodel_hh(5)	ts(96)_states(20)	-143169	-152225	202	6456
daytimemodel_hh(5)	ts(96)_states(25)	-139531	-151509	198	6410
daytimemodel_hh(5)	ts(96)_states(30)	-135909	-147916	202	6518
daytimemodel_hh(5)	ts(96)_states(40)	-121243	-132364	196	6294
daytimemodel_hh(5)	ts(96)_states(50)	-113322	-124996	184	5928
daytimemodel_hh(5)	ts(96)_states(60)	-107835	-120324	191	6156
seasonmodel_hh(5)	spring_states(5)	-8903	-10533	113	3829
seasonmodel_hh(5)	spring_states(10)	-8179	-9685	102	3521
seasonmodel_hh(5)	spring_states(15)	-7904	-9253	101	3369
seasonmodel_hh(5)	spring_states(20)	-7788	-9326	102	3376
seasonmodel_hh(5)	spring_states(25)	-7753	-9466	101	3397
seasonmodel_hh(5)	spring_states(30)	-7624	-9629	101	3434
seasonmodel_hh(5)	spring_states(40)	-7663	-9227	100	3339
seasonmodel_hh(5)	spring_states(50)	-7335	-9678	98	3369
seasonmodel_hh(5)	spring_states(60)	-7313	-10465	101	3468
seasonmodel_hh(5)	summer_states(5)	-6359	-7571	80	2630
seasonmodel_hh(5)	summer_states(10)	-5935	-7305	74	2508
seasonmodel_hh(5)	summer_states(15)	-5756	-7255	76	2539
seasonmodel_hh(5)	summer_states(20)	-5673	-7281	74	2504
seasonmodel_hh(5)	summer_states(25)	-5503	-7307	75	2515
seasonmodel_hh(5)	summer_states(30)	-5943	-7393	79	2572
seasonmodel_hh(5)	summer_states(40)	-5659	-7393	76	2523
seasonmodel_hh(5)	summer_states(50)	-5248	-7502	76	2518
seasonmodel_hh(5)	summer_states(60)	-5439	-7528	76	2539
seasonmodel_hh(5)	fall_states(5)	-9158	-10689	120	3945
seasonmodel_hh(5)	fall_states(10)	-8415	-9987	107	3576
seasonmodel_hh(5)	fall_states(15)	-8182	-10076	106	3524
seasonmodel_hh(5)	fall_states(20)	-7707	-9863	107	3535
seasonmodel_hh(5)	fall_states(25)	-7757	-9840	105	3523
seasonmodel_hh(5)	fall_states(30)	-7848	-10015	105	3524
seasonmodel_hh(5)	fall_states(40)	-7517	-9878	104	3485
seasonmodel_hh(5)	fall_states(50)	-7275	-10297	105	3546
seasonmodel_hh(5)	fall_states(60)	-7173	-10213	108	3614
seasonmodel_hh(5)	winter_states(5)	-15334	-19028	286	9778
seasonmodel_hh(5)	winter_states(10)	-12324	-13855	121	4032
seasonmodel_hh(5)	winter_states(15)	-11653	-13584	119	3935
seasonmodel_hh(5)	winter_states(20)	-11447	-13064	116	3847
seasonmodel_hh(5)	winter_states(25)	-11364	-13446	117	3919
seasonmodel_hh(5)	winter_states(30)	-11242	-13413	118	3935
seasonmodel_hh(5)	winter_states(40)	-11116	-13396	117	3879
seasonmodel_hh(5)	winter_states(50)	-10874	-13428	116	3922
seasonmodel_hh(5)	winter_states(60)	-10730	-14074	122	4025

FIG. 6. Model evaluations for household Nr.5

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